

Environmental Belief System Networks and Pro-Environmental Behavior

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Introduction

Explaining Pro-Environmental Behavior

Belief System Networks

Data and Measures

I use data from the 2020 wave of the International Social Survey Programme (ISSP) environment module, which surveyed 44,100 respondents across 28 countries. The module repeats a set of environmental attitude and behavior items that have been fielded since 1993, and it allows the belief network approach to be examined across a wide range of national contexts.

The ISSP does not include established measures such as the New Ecological Paradigm (NEP) or the Connectedness to Nature Scale (CNS), so I construct belief measures from the module's attitude items. Using exploratory graph analysis I identify four belief dimensions and score each as the mean of its items, oriented so that higher values indicate more pro-environmental positions. Environmental commitment combines concern about the environment with willingness to pay higher prices and taxes, willingness to cut one's standard of living, and a personal norm to act ($\alpha = 0.75$). Ecological worldview captures low environmental and climate change skepticism and is the closest available analog to the NEP ($\alpha = 0.76$). Perceived threat is the mean of seven items rating the danger posed by specific environmental hazards ($\alpha = 0.80$). Nature affinity combines enjoyment of nature with the frequency of outdoor leisure and is a limited analog to the CNS ($\alpha = 0.61$). The supplemental materials describe the scale construction.

I also include three political and economic beliefs as single-item measures. Ideology is the left-right position of the party the respondent reports voting for, which is unavailable in 10 countries. Support for private enterprise

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is agreement that private enterprise is the best way to solve the country's economic problems, and support for government redistribution is agreement that it is the responsibility of the government to reduce the differences in income between people with high and low incomes. Both are measured on five-point scales with higher values indicating more agreement. The two items are only weakly related, so I include them as separate nodes rather than combining them into a scale.

The dependent variable is pro-environmental behavior, which I separate into public and private behavior. Public behavior is a count of four collective actions, membership in an environmental group, signing a petition, donating money, and taking part in a protest. Private behavior is a count of two household actions, recycling and avoiding environmentally harmful products, each coded as done always or often. Exploratory graph analysis of the six items recovers this public and private grouping.

Results

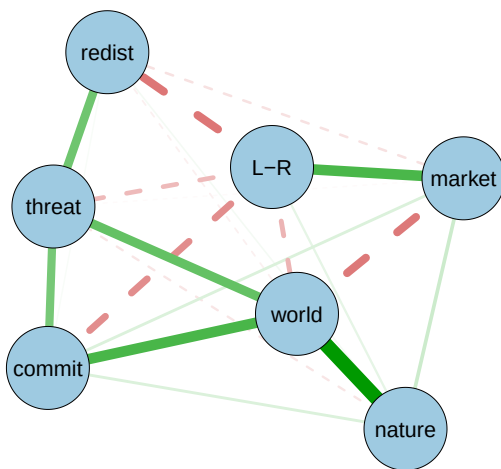
I estimate a regularized partial correlation network of the belief and behavior measures for each of the 28 countries and for the pooled sample of all countries, using the graphical lasso with EBIC model selection. Each edge represents the association between two nodes after controlling for all other nodes in the network. To measure how close each belief is to behavior, I compute the shortest path from each belief node to the public and private behavior nodes using Dijkstra's algorithm, with the length of each edge defined as the inverse of its absolute weight. A shorter path indicates a belief more closely connected to behavior.

Figure 1 shows the pooled network with and without the behavior nodes. In the belief network, ecological worldview is the most connected node, with its strongest ties to nature affinity (.32) and environmental commitment (.23). Ideology is linked to the two economic beliefs, positively with support for private enterprise (.23) and negatively with support for government redistribution (-.17). The political and economic beliefs connect to the environmental beliefs mainly through negative ties with ecological worldview and environmental commitment and a positive tie between redistribution and perceived threat. Once behavior is added, environmental commitment has the strongest tie to public behavior (.25) and also connects to private behavior (.18), while ecological worldview and nature affinity connect to both. Ideology has a negative tie to public behavior (-.13), and perceived threat and the economic beliefs have little or no direct connection to either type of behavior. In the pooled network environmental commitment has the shortest path to public behavior (3.93) and nature affinity the shortest path to private behavior (5.26), with environmental commitment close behind (5.51).

However, the pooled network masks differences across countries. Figure 2 shows the networks for four countries that span the range of results. In Switzerland, the typical case, environmental commitment is closest to behavior in 100 percent of bootstrap resamples. In the United States nature affinity has the shortest path to behavior (60 percent of resamples), but environmental commitment and ideology are not far behind. India is the clearest case where a belief other than environmental commitment is closest, with nature affinity holding in 97 percent of resamples. In Australia ideology, environmental commitment, and ecological worldview are close to a three-way tie.

Figure 1: Pooled belief network across all countries, without (left) and with (right) the public and private behavior nodes. Blue nodes are the belief measures (commit = environmental commitment, world = ecological worldview, threat = perceived threat, nature = nature affinity, L-R = left-right ideology, market = private enterprise, redistrib = government redistribution), orange nodes are the behavior indices. Solid edges are positive regularized partial correlations and dashed edges negative, with thicker edges indicating stronger associations. Node positions and edge widths are on a common scale across the two panels.

Beliefs



Beliefs and behavior

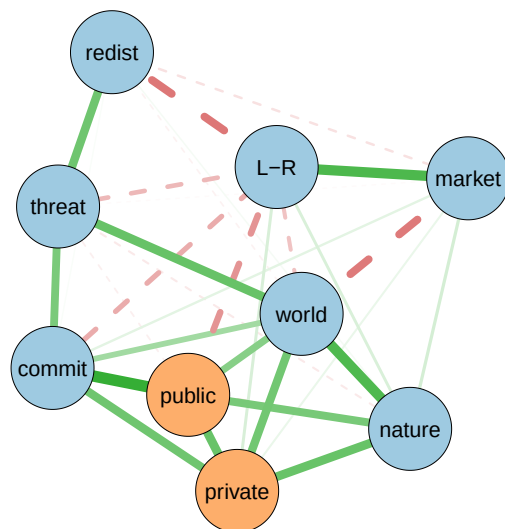


Table 1 reports the mean rank of each belief across countries, where a rank of one identifies the belief closest to behavior in a given country. Environmental commitment has the shortest mean path to behavior in 24 of 28 countries and a mean rank of 1.29, followed by ecological worldview (2.71) and nature affinity (3.21). Perceived threat and ideology are further from behavior, and the two economic beliefs are the most peripheral. A nonparametric bootstrap with 1,000 resamples per country shows that this ordering is stable. The belief identified as closest holds in a majority of resamples in 27 of 28 countries, so the cross-country pattern is not an artifact of a few unstable networks.

Table 1: Mean rank of each Study 2 belief across the 28 ISSP countries (1 = closest to behavior). ‘Closest in’ counts the countries where the belief has the shortest mean path to behavior, and ‘bootstrap-supported’ the subset where it holds in a majority of 1,000 resamples.

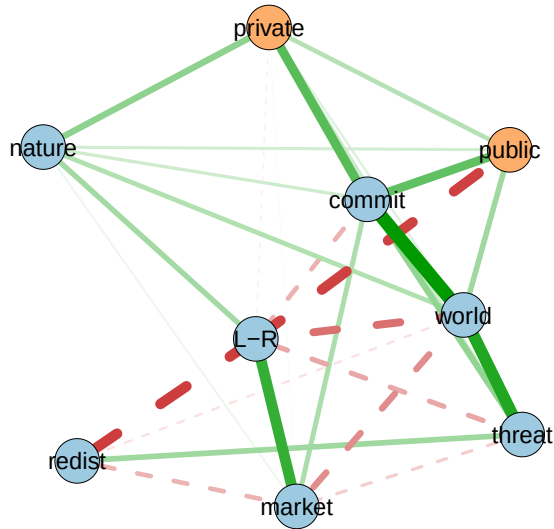
Belief	Mean rank	Closest in	Bootstrap-supported
Environmental commitment	1.29	24	24
Ecological worldview	2.71	1	0
Nature affinity	3.21	3	3
Perceived threat	4.29	0	0
Ideology (left-right)	4.61	1	0
Private enterprise	5.18	0	0
Government redistribution	5.79	0	0

Figure 3 shows the shortest path from each belief to each type of behavior across the countries. Environmental commitment has the shortest average path to both public (6.2) and private behavior (6.3). Its confidence interval does not overlap with that of any other belief for private behavior, and it overlaps only slightly with ecological worldview for public behavior. Ecological worldview is the next closest belief to public behavior (8.9), while nature affinity is the next closest to private behavior (9.7). The political and economic beliefs are the furthest from both types of behavior, although ideology varies widely across the countries where it is available.

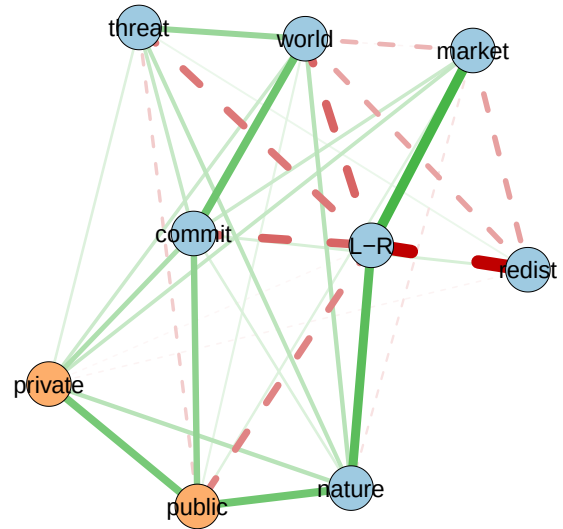
As noted, the belief that combines concern, willingness to pay, and a personal norm to act sits closest to behavior in most countries, while ecological worldview, perceived threat, and the political and economic beliefs are further away. This is likely because beliefs tied to personal commitment are more proximate to action than beliefs about how the natural world works or about the role of government in the economy.

Figure 2: Belief networks for four illustrative Study 2 countries. Blue nodes are the belief measures (commit = environmental commitment, world = ecological worldview, threat = perceived threat, nature = nature affinity, L-R = left-right ideology, market = private enterprise, redist = government redistribution), orange nodes are the public and private behavior indices. Solid edges are positive regularized partial correlations and dashed edges negative, with thicker edges indicating stronger associations.

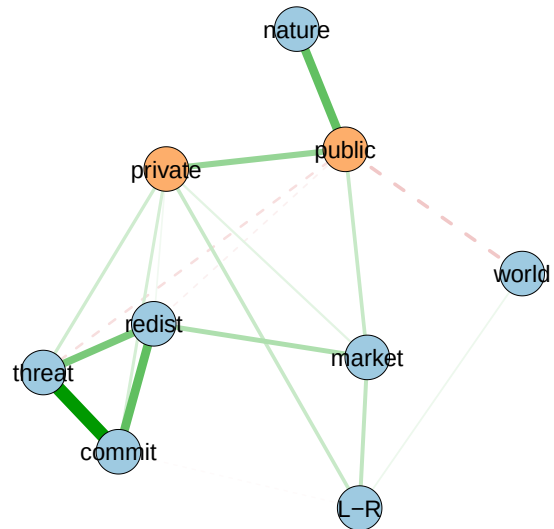
Switzerland (n = 4280)



United States (n = 1847)



India (n = 1421)



Australia (n = 1147)

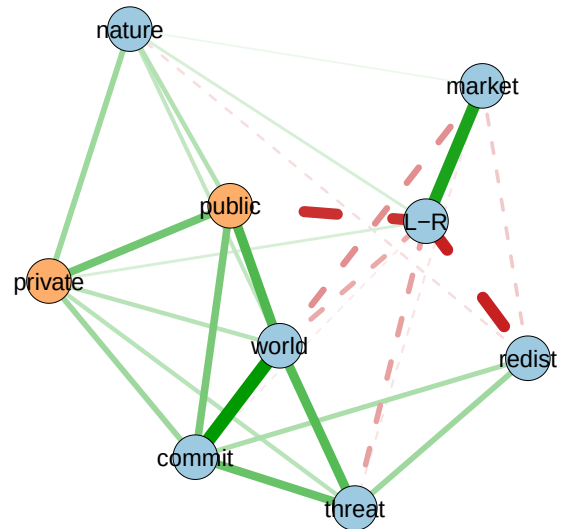
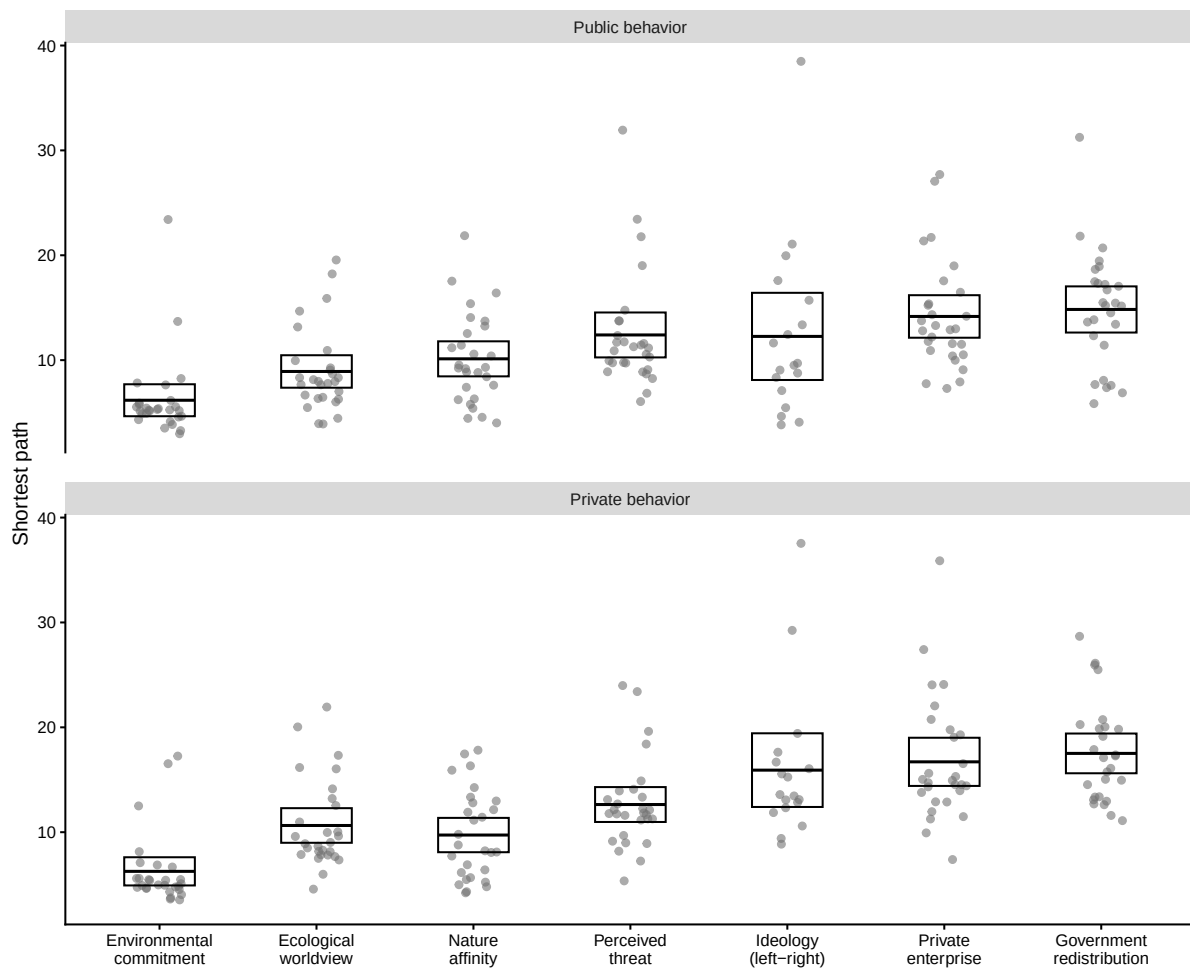


Figure 3: Shortest path from each belief to public and private behavior across the Study 2 countries. Lower values indicate a belief closer to behavior. Points are individual countries, horizontally jittered. The centerline of each box is the mean across countries and the top and bottom edges are the bounds of the 95 percent confidence interval of the mean. Ideology is available in 18 countries, and government redistribution is disconnected from behavior in Taiwan and omitted there.



References